

Glucose Result Release Advisor

Everything the system can ask, find, say and decide, with a worked case for each. Generated by running the knowledge base, not written from memory, so it cannot drift from the code.

14 questions · 15 faults · 8 actions · 30 worked cases

Every case in part 7 was run through the live knowledge base. The inputs are what you type, and the outputs are what the system actually produced. If a case here disagrees with what you see on screen, something is wrong.

1. Running it

To do this	Do this	You need
Just use the system	Double-click webindex.html	A browser. Nothing else.
Edit the rules and see changes on refresh	Double-click run.bat , then open the address it opens	Python 3
Refresh the page after editing kb/	Double-click build.bat	Python 3
Run the test suite	node test/cases.js	Node.js 16+

The footer of the page tells you which knowledge source is in use: kb/*.pl, loaded live when served, or the copy embedded in this page when opened from disk. Both reason identically. Nothing is sent anywhere, so it works with the network off.

2. Every question it can ask

Asked in this order. The system stops as soon as no remaining question could change the action, so most consultations ask fewer than all 14.

Rank	The question	Answers, and what each records	Fault it can establish
1	Does the label on the tube match the request form? always asked	Yes, it matches → <i>nothing recorded</i> No, it does not → label_mismatch(s).	label does not match the request
2	Any doubt about what was typed into the system? always asked	No doubt → <i>nothing recorded</i> Yes, there is doubt → transcription_doubt(s).	entry may have been typed wrongly
3	How many hours had the patient fasted? only if glucose was requested	a number in hours	patient was not fasting
3	Was the blood drawn from the arm with the drip? always asked	No → <i>nothing recorded</i> Yes → drip_arm(s).	drawn from the drip arm
4	How does the sample look? always asked	Normal → <i>nothing recorded</i> Haemolysed → sample_state(s, haemolysed). Clotted → sample_state(s, clotted). Lipaemic → sample_state(s, lipaemic).	sample haemolysed, sample clotted, sample lipaemic
4	Which tube was the glucose drawn into? only if glucose was requested	Fluoride oxalate → tube(s, fbs, fluoride). EDTA → tube(s, fbs, edta). Plain → tube(s, fbs, plain). Heparin → tube(s, fbs, heparin).	the wrong tube was used

Rank	The question	Answers, and what each records	Fault it can establish
4	Which tube was the HbA1c drawn into? only if HbA1c was requested	EDTA → tube(s, hba1c, edta). Fluoride oxalate → tube(s, hba1c, fluoride). Plain → tube(s, hba1c, plain). Heparin → tube(s, hba1c, heparin).	the wrong tube was used
4	Which method measures HbA1c here? only if HbA1c was requested	Ion-exchange HPLC → hba1c_method(s, hplc). Immunoassay → hba1c_method(s, immunoassay). Boronate affinity → hba1c_method(s, boronate). Enzymatic → hba1c_method(s, enzymatic). Not recorded → <i>nothing recorded</i>	—
4	How many hours before the plasma was separated? only if glucose was requested	a number in hours	sample sat too long before separation
5	Did the quality control for this run pass? always asked	Passed → <i>nothing recorded</i> Out of range → qc_status(s, fbs, out_of_range).	quality control out of range
5	Is the reagent lot within its expiry date? always asked	Within expiry → <i>nothing recorded</i> Expired → reagent_lot(s, fbs, expired).	reagent lot past its expiry
5	Is the calibration current? always asked	Current → <i>nothing recorded</i> Overdue → calibration(s, fbs, overdue).	calibration overdue
5	Did the analyser raise a flag on this sample? always asked	None → <i>nothing recorded</i> Probe clot → analyser_flag(s, probe_clot). Carryover → analyser_flag(s, carryover).	probe clot flagged by the analyser, carryover from the previous sample
5	Was the analyser cleaned as scheduled? always asked	Yes → <i>nothing recorded</i> No → instrument_clean(s, no).	analyser not cleaned

Rank is the best action that question could establish. Questions are asked in rank order, which is why the system can stop early: once a fault is found at rank *R*, nothing still unasked could beat it.

3. Every fault it can find

The family decides whether the patient is bled again. That is the whole system.

What it is called	In Prolog	What it does to the result
MACHINE — THE BLOOD IS INNOCENT, SO RE-RUN THE SAME TUBE		
quality control out of range	qc_out	glucose: unpredictable – the run was not in control, so the direction of the error is unknown HbA1c: unpredictable – the run was not in control, so the direction of the error is unknown
reagent lot past its expiry	reagent_expired	glucose: unpredictable – an expired lot is outside the conditions it was validated for, so the result cannot be assumed reliable
calibration overdue	calibration_overdue	glucose: unpredictable – an uncalibrated run cannot be trusted in either direction
probe clot flagged by the analyser	probe_clot	<i>nothing recorded</i>
carryover from the previous sample	carryover	glucose: pushes it up – a very high sample just before this one can carry into the probe
analyser not cleaned	not_cleaned	glucose: unpredictable – residue in the path contaminates the reading
SAMPLE — THE SPECIMEN CANNOT GIVE A SOUND RESULT, SO FRESH BLOOD		
sample haemolysed	haemolysed	<i>nothing recorded</i>
sample clotted	clotted	<i>nothing recorded</i>
sample lipaemic	lipaemic	glucose: unpredictable – turbidity interferes with the reading, by an amount that depends on the analyser and the method
the wrong tube was used	wrong_tube	glucose: pushes it down – without fluoride the cells consume the glucose faster still
sample sat too long before separation	delayed	glucose: pushes it down – the cells go on eating the glucose in the tube, a reported average of about 5 to 7 per cent an hour
COLLECTION — FRESH BLOOD, AND THE PATIENT HAS TO BE TOLD SOMETHING		
patient was not fasting	not_fasting	glucose: pushes it up – the patient had eaten, so this is not a fasting value Tell the patient: fast for at least 8 hours before the next draw - no food and no sugary drinks; plain water is allowed
drawn from the drip arm	drip_arm	glucose: pushes it up – a sample drawn near a running drip carries the drip fluid with it Tell the patient: draw from the opposite arm, away from any running drip
IDENTITY — DEPENDS: A LABEL MISMATCH IS URGENT, A TYPING DOUBT NEEDS NO BLOOD		
label does not match the request	label_mismatch	<i>nothing recorded</i>
entry may have been typed wrongly	transcription_doubt	<i>nothing recorded</i>

Nothing recorded is a gap in the knowledge base, not a finding that the fault has no effect. The system says so rather than guessing, and those gaps are what the domain expert still has to fill.

4. Every action it can reach

When more than one is possible, the lowest rank wins.

Rank	What it says	The reason it gives	Requires these families absent
1	Take a fresh sample NOW, and tell the senior	the label does not match the request, so this may be the wrong patient	—
2	Correct the entry - no new blood needed	the result itself is sound, so check what was typed against the analyser	sample, collection, machine
3	Take a fresh sample, and tell the patient what to do	the sample was not collected correctly, and the patient has to be told what to do differently	—
4	Take a fresh sample	this specimen cannot give a sound result	—
5	Re-run the same sample	the analyser was at fault, so the blood in the tube is still good - but the analyser has to be put right before the sample is run again	sample, collection
6	Call the senior	(from the rule that fired)	machine, sample, collection, identity
7	Release, but add a comment	no fault found, but the change from the last result is large	machine, sample, collection, identity
8	Release the result	no fault found and the result is plausible	machine, sample, collection, identity

The last column is why the system will not stop early on some answers. An action that rests on the absence of a fault is only trusted once every question that could establish one has been asked. Not asking is not the same as finding nothing.

5. Every number it checks against

What it is	Test	Value	In Prolog
Normal range	glucose	70 to 99	reference_range
Normal range	HbA1c	4.0 to 5.6	reference_range
Below this, no patient survives	glucose	20	survivable_low
Above this, no patient survives	glucose	800	survivable_high
Below the assay range	HbA1c	3.0	impossible_low
Above the assay range	HbA1c	20.0	impossible_high
Critically low, telephone	glucose	50	critical_low
Critically high, telephone	glucose	400	critical_high
Largest change from the last result	glucose	100	delta_limit
Largest change from the last result	HbA1c	2.0	delta_limit
Longest wait before separation	glucose	1	max_delay_hours
The correct tube	glucose	fluoride	proper_tube
The correct tube	HbA1c	edta	proper_tube
Shortest acceptable fast	glucose	8	min_fasting_hours

6. The four things it can say about a number

Separate from what to do with the tube. These decide what the report can say about the result.

It says	When	To see it
Consistent	The fault pushes the result the way it actually went, so it could be the cause	Glucose 61 , separated after 4 hours
Does not fit	The fault pushes the opposite way, so it is not what caused this	Glucose 350 , separated after 4 hours
Cannot say	The fault affects this test in an unpredictable direction	Glucose 268 , quality control out of range
Not recorded	Nothing records what this fault does to this test	Glucose 350 , sample haemolysed
Probably real	Abnormal, and nothing found pushes it that way or is unpredictable	Glucose 350 , separated after 4 hours

7. A worked case for every fault

Answer everything with the good option except where the third column says otherwise. The last two columns are what the system actually produced when this manual was generated.

Fault	Type in	Answer differently	You should see
MACHINE			
quality control out of range	Glucose 61	Did the quality control for this run pass? → Out of range	Re-run the same sample asked 8 of 12
reagent lot past its expiry	Glucose 210	Is the reagent lot within its expiry date? → Expired	Re-run the same sample asked 9 of 12
calibration overdue	Glucose 118	Is the calibration current? → Overdue	Re-run the same sample asked 10 of 12
probe clot flagged by the analyser	Glucose 88	Did the analyser raise a flag on this sample? → Probe clot	Re-run the same sample asked 11 of 12
carryover from the previous sample	Glucose 402	Did the analyser raise a flag on this sample? → Carryover	Re-run the same sample asked 11 of 12 Flagged critically high
analyser not cleaned	Glucose 97	Was the analyser cleaned as scheduled? → No	Re-run the same sample asked 12 of 12
SAMPLE			
sample haemolysed	Glucose 150	How does the sample look? → Haemolysed	Take a fresh sample asked 5 of 12
sample clotted	Glucose 150	How does the sample look? → Clotted	Take a fresh sample asked 5 of 12
sample lipaemic	Glucose 150	How does the sample look? → Lipaemic	Take a fresh sample asked 5 of 12
the wrong tube was used	Glucose 61	Which tube was the glucose drawn into? → Plain	Take a fresh sample asked 6 of 12
sample sat too long before separation	Glucose 61	How many hours before the plasma was separated? → 4	Take a fresh sample asked 7 of 12
COLLECTION			
patient was not fasting	Glucose 140	How many hours had the patient fasted? → 3	Take a fresh sample, and tell the patient what to do asked 3 of 12
drawn from the drip arm	Glucose 188	Was the blood drawn from the arm with the drip? → Yes	Take a fresh sample, and tell the patient what to do asked 4 of 12
IDENTITY			
label does not match the request	Glucose 310	Does the label on the tube match the request form? → No, it does not	Take a fresh sample NOW, and tell the senior asked 1 of 12
entry may have been typed wrongly	Glucose 102	Any doubt about what was typed into the system? → Yes, there is doubt	Correct the entry - no new blood needed asked 12 of 12

8. Edge cases and combinations

The interesting ones: clean runs, impossible numbers, two faults at once, and the cases where the reasoning matters more than the action.

The case	Type in	Answer differently	You should see
A clean glucose	Glucose 94	<i>nothing, answer it all clean</i>	Release the result asked 12 of 12
A clean glucose and HbA1c together	Glucose 94 , HbA1c 5.3	<i>nothing, answer it all clean</i>	Release the result asked 14 of 14
An HbA1c on its own	HbA1c 5.3	<i>nothing, answer it all clean</i>	Release the result asked 11 of 11
Critically low, but nothing wrong	Glucose 48	<i>nothing, answer it all clean</i>	Release the result asked 12 of 12 Flagged critically low
		Under Why :	Probably real. this result is low and nothing found in the run pushes it that way, so treat the low value as the patient's own until proved otherwise
Critically high, but nothing wrong	Glucose 430	<i>nothing, answer it all clean</i>	Release the result asked 12 of 12 Flagged critically high
		Under Why :	Probably real. this result is high and nothing found in the run pushes it that way, so treat the high value as the patient's own until proved otherwise
Below what a patient survives	Glucose 12	<i>nothing, answer it all clean</i>	Call the senior asked 12 of 12 Flagged critically low
HbA1c below the assay floor	HbA1c 1.5	<i>nothing, answer it all clean</i>	Call the senior asked 11 of 11
HbA1c above the assay ceiling	HbA1c 22	<i>nothing, answer it all clean</i>	Call the senior asked 11 of 11
Two faults: specimen beats machine	Glucose 61	How does the sample look? → Haemolysed Did the quality control for this run pass? → Out of range	Take a fresh sample asked 10 of 12 Also: fix quality control out of range first Offers the extra questions first
		Under Why :	Cannot say. quality control out of range. this fault affects the test, but not in a predictable direction, so this result cannot be judged either way Not recorded. sample haemolysed. the knowledge base does not record what this fault does to this test, so this result cannot be judged against it
Two faults: identity beats everything	Glucose 61	Does the label on the tube match the request form? → No, it does not How does the sample look? → Haemolysed	Take a fresh sample NOW, and tell the senior asked 1 of 12 Offers the extra questions first
		Under Why :	Not recorded. label does not match the request. the knowledge base does not record what this fault does to this test, so this result cannot be judged against it
A delay cannot explain a HIGH result	Glucose 350	How many hours before the plasma was separated? → 4	Take a fresh sample asked 7 of 12 Offers the extra questions first

The case	Type in	Answer differently	You should see
		Under Why :	Does not fit. sample sat too long before separation. this fault pushes the result down, but this result is high, so it is not what caused that Probably real. this result is high and nothing found in the run pushes it that way, so treat the high value as the patient's own until proved otherwise
A delay CAN explain a low result	Glucose 61	How many hours before the plasma was separated? → 4	Take a fresh sample asked 7 of 12 Offers the extra questions first
		Under Why :	Consistent. sample sat too long before separation. this fault pushes the result down, and this result is low, so it could well be the cause
A typing doubt beside a bad specimen	Glucose 94	Any doubt about what was typed into the system? → Yes, there is doubt How does the sample look? → Haemolysed	Take a fresh sample asked 5 of 12 Offers the extra questions first
HbA1c in the glucose tube	Glucose 94 , HbA1c 5.3	Which tube was the HbA1c drawn into? → Fluoride oxalate	Take a fresh sample asked 7 of 14 Offers the extra questions first
		Under Why :	Method missing. the HbA1c method was not recorded, and whether a fault affects an HbA1c depends on it
Accepting the offer to check the analyser	Glucose 61	Which tube was the glucose drawn into? → Plain Was the analyser cleaned as scheduled? → No	Take a fresh sample asked 11 of 12 Also: fix analyser not cleaned first Offers the extra questions first
		Under Why :	Consistent. the wrong tube was used. this fault pushes the result down, and this result is low, so it could well be the cause Cannot say. analyser not cleaned. this fault affects the test, but not in a predictable direction, so this result cannot be judged either way

One case you cannot reach by answering questions. The delta check needs the patient's previous result, which is an optional field on the first screen, not a question. Open *Add HbA1c or a previous result*, enter glucose **88** with a last glucose of **240**, and answer everything clean. You should see **Release, but add a comment**: no fault found, but the change from the last result is large.

What the system never does: it does not say what a result means for a patient and it does not name a disease. It decides whether a number is fit to leave the laboratory.

Prakasan R. · 224152U · University of Moratuwa